SHANGHAI JIAO TONG UNIVERSITY

StockCL: Selective Contrastive Learning for Stock Trend Forecasting via Learnable Concepts

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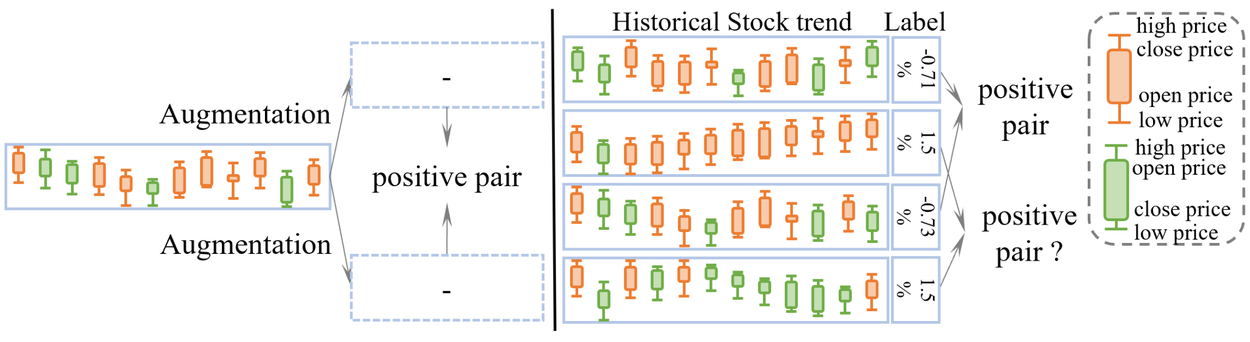
Problem & Motivation

Stock Trend Forecasting

- Limited data leads to easy overfitting of deep learning models, hampering their generalization and backtest performance.
- How to provide additional supervision and prevent overfitting?

Contrastive Learning (CL)

- CL enhances the generalization of deep learning models by introducing additional supervision signals.
- CL generates positive and negative pairs, pulling representations of positives closer and pushing negatives further to learn robust representations.



Challenging issues

- Augmentations: no intuitive data perturbation methods. Some methods like adding Gaussian noise can distort underlying patterns.
- Select contrastive samples based on label similarity like supervised contrastive learning: sub-optimal. Samples with similar labels (i.e., price change rate) can exhibit distinct input features (i.e., historical trends).

Motivation

- Stock trends are driven by latent concepts such as market sentiment, capital flows, and broader market conditions, etc.
- Key insight: samples sharing the same concept should have closer representations.
- We identify latent concepts based on sample input features and use them to filter noisy positive pairs generated solely by label similarity.
- Our approach constructs positive pairs by incorporating both label and concept information.

Problem Definition

Stock Features

- At time step t, the features of a stock s: $\mathbf{x}_t^{(s)} \in \mathbb{R}^D$.

Stock Trend

- The relative change rate of the daily stock price

$$y_t^{(s)} = \frac{Price_{t+1}^{(s)} - Price_t^{(s)}}{Price_t^{(s)}} \in \mathbb{R}$$

Problem Definition: Stock Trend Forecasting

- To learn a forecasting model $f(\cdot)$ to predict $y_t^{(s)}$ given a collection of daily features on the past time window

$$\mathbf{X}_t^{(s)} = \left[\mathbf{x}_{t-T+1}^{(s)}, \dots, \mathbf{x}_{t-1}^{(s)}, \mathbf{x}_t^{(s)}\right] \in \mathbb{R}^{T*D}.$$

- The forecasting model $f(\cdot)$ can be split into an encoder $f(\cdot)$ and a predictor $p(\cdot)$ with representations $\mathbf{z}_i$

$$\mathbf{z}_t^{(s)} = f\left(\mathbf{X}_t^{(s)}\right) \in \mathbb{R}^d, \hat{y}_t^{(s)} = p(\mathbf{z}_t^{(s)})$$

Training samples

- $\mathcal{T} = \left\{\left(\mathbf{X}_t^{(s)}, \hat{y}_t^{(s)}\right)\right\}_{i=1}^{N}$ across all stocks and all time steps.

StockCL framework

Concept Pool

- Concept representations. $\mathbf{C} = \{\mathbf{c}_1, \mathbf{c}_2 \dots, \mathbf{c}_M\}, \mathbf{c}_k \in \mathbb{R}^d$.
- Concept matching. Matching score between sample $\mathbf{z}_i$ and the kth concept:

$$\alpha_{ik} = \frac{\exp(\mathbf{z}_i^T \mathbf{c}_k)}{\sum_{m=1}^{M} \exp(\mathbf{z}_i^T \mathbf{c}_m)},$$

The retrieved concept index: $d_i = argmax_k \alpha_{ik}$.

- Concept loss. Ideally, **each learned concept should be in the cluster center of the samples sharing this concept.** At the same time, **the learned concepts should be discriminative on the samples to make sure that each sample's matched concept is the dominant one.** Hence, the concept loss contains two parts: L_{clr} and L_{dis}.

$$L_{\text{clr}} = \sum_{k=1}^{M} \sum_{i \in J} \mathbb{I}(d_i = k) ||\mathbf{c}_k - \mathbf{z}_i||_2^2,$$

$$L_{\text{dis}} = \sum_{i \in J} \max(0, ||\mathbf{z}_i - \mathbf{c}_{d_i}||_2^2 - ||\mathbf{z}_i - \mathbf{c}_{d_i^{(2)}}||_2^2 + \delta),$$

where $d_i^{(2)} = argmax_{k, k \neq d_i} \alpha_{ik}$ is the concept index with the second highest matching score for z_i.

- The concept loss can be denoted as $L_1 = L_{\text{clr}} + L_{\text{dis}}$.

Concept-aware Contrastive Pair Selector

- Contrastive pair selection. **We define positive pairs as those with similar labels and sharing the same concept.**

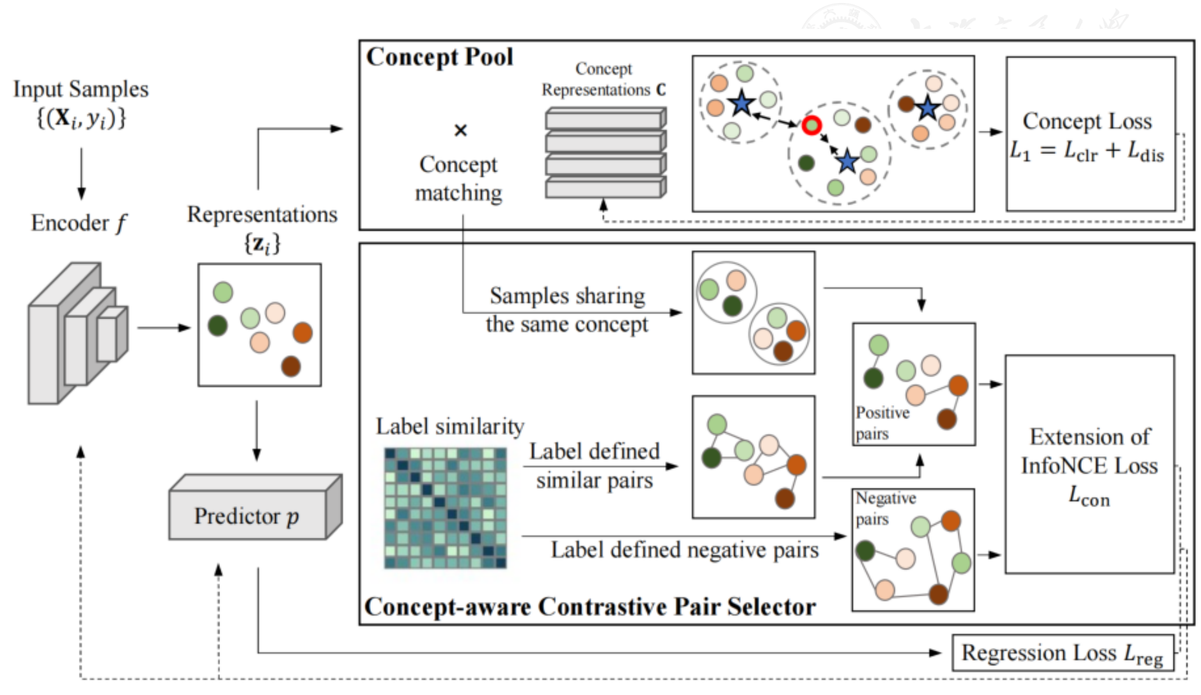
$$\mathcal{S}_i^{pos} = \{(X_j, y_j) \in \mathcal{A} | d_i = d_j, d(y_i, y_j) < \epsilon^{pos}\}$$

$$\mathcal{S}_i^{neg} = \{(X_j, y_j) \in \mathcal{A} | d(y_i, y_j) > \epsilon^{neg}\}$$

Extension of InfoNCE

- $L_{con} = \sum_{i \in J} L_{con}^i = \sum_{i \in J} -\frac{1}{|\mathcal{S}_i^{pos}|} \sum_{j \in \mathcal{S}_i^{pos}} \beta_{ij} log \frac{\exp(z_i^T z_j / \tau)}{\sum_{k \in \mathcal{S}_i^{neg}} \exp(\frac{z_i^T z_k}{\tau})}$

  where β_{ij} is the similarity between sample i and j.
- The loss to train the encoder and the predictor is $L_2 = L_{reg} + \lambda L_{con}$. $L_{reg} = \sum_{i \in J} (\hat{y}_i - y_i)^2$.



Experiments

Datasets

- CSI300
- CSI500

	Training			Validation			Test		
	#days	#stocks	#samples	#days	#stocks	#samples	#days	#stocks	#samples
CSI300	1684	505	478K	456	386	128K	868	451	253K
CSI500	1684	1058	785K	456	677	205K	868	810	418K

Base Models

- GRU
- ALSTM
- Localformer
- GAT

Metric

- IC
- ICIR
- RankIC
- RankICIR

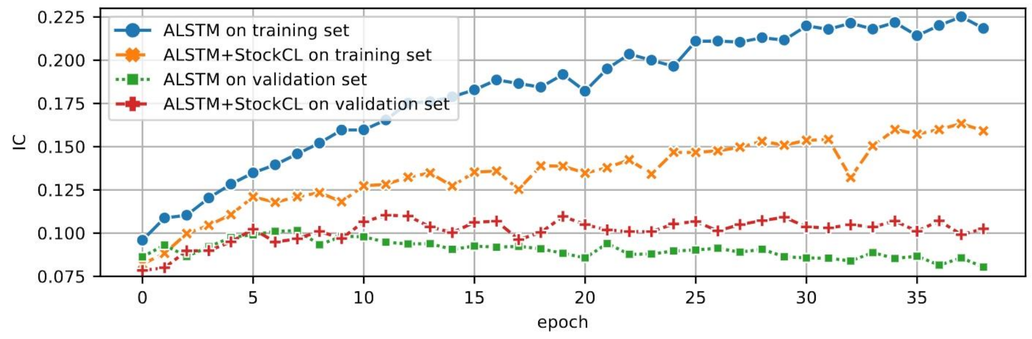
Baselines

- ConAlign
- CRGA
- RNC

Model	Method	CSI300 IC	CSI300 ICIR	CSI300 RankIC	CSI300 RankICIR	CSI500 IC	CSI500 ICIR	CSI500 RankIC	CSI500 RankICIR
GRU	-	0.055	0.453	0.056	0.474	0.053	0.584	0.054	0.603
	ConAlign	0.058	0.460	0.052	0.441	0.057	0.596	0.048	0.524
	CRGA	0.053	0.415	0.054	0.438	0.052	0.531	0.050	0.515
	RNC	0.059	0.473	0.053	0.451	0.057	**0.599**	0.052	0.567
	StockCL	**0.063**	**0.533**	**0.056**	**0.497**	**0.059**	0.583	**0.057**	**0.604**
ALSTM	-	0.053	0.365	0.053	0.362	0.052	0.533	0.056	0.620
	ConAlign	0.055	0.397	0.054	0.380	0.053	0.540	0.050	0.508
	CRGA	0.050	0.394	0.048	0.378	0.048	0.462	0.041	0.398
	RNC	0.058	0.471	0.056	0.474	0.053	0.526	0.055	0.599
	StockCL	**0.066**	**0.546**	**0.056**	**0.502**	**0.059**	**0.604**	**0.057**	**0.628**
Local-former	-	0.052	0.460	0.053	**0.488**	0.048	0.472	0.050	0.532
	ConAlign	0.053	0.432	0.048	0.406	0.050	0.519	**0.053**	**0.565**
	CRGA	0.033	0.220	0.028	0.178	0.027	0.261	0.029	0.273
	StockCL	**0.059**	**0.471**	**0.054**	0.455	**0.053**	**0.565**	0.048	0.557
GAT	-	0.054	0.434	0.047	0.398	0.048	0.479	0.048	0.485
	ConAlign	0.053	0.414	0.049	0.408	0.051	0.524	0.050	0.524
	CRGA	0.017	0.107	0.021	0.128	0.022	0.191	0.030	0.299
	RNC	0.055	0.424	0.050	0.407	0.051	0.503	0.052	**0.561**
	StockCL	**0.057**	**0.441**	**0.052**	**0.433**	**0.056**	**0.563**	**0.053**	0.552

Analysis

- StockCL surpasses on the four stock trend forecasting models by 6%, 18%, 5%, 11% on the two datasets in average and generally outperforms the three baseline methods.



Conclusion & Contribution

- We propose StockCL, a novel supervised contrastive learning framework for stock trend forecasting to mitigate data hungry and overfitting issues. StockCL is model-agnostic.
- StockCL consists concept pool and concept-aware contrastive pair selector components. The former learns concept representations in the embedding space. The latter constructs reliable contrastive pairs based on samples' belonging concepts and label similarity and applies contrastive learning.
- We conduct extensive experiments on two real-world datasets and four stock trend forecasting models. StockCL outperforms state-of-the-art supervised contrastive learning approaches and consistently improves the models' stock trend forecasting performance from 5% to 18%.

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